EY TAX ADVISORY & REGULATORY AUTOMATION

TAX LAW DELTA ANALYZER

Deterministic Regulatory State-Diffing and Local Context Refinement Engine

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PRESENTER

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EVALUATION METADATA

Enterprise AI POC • Token Optimization • Zero-Hallucination Compliance

PROBLEM STATEMENT

The Operational Bottleneck in Corporate Tax Advisory

The Manual Highlighter Struggle

- ⚠ **Consultants spend hours cross-referencing new 50+ page PDFs against prior codifications.**
- ⚠ **Human fatigue leads to missed clause updates, sunset provisions, and miscalculated effective dates.**

The Commercial Cloud LLM Dilemma

- ☁ **Ingesting dual 50-page PDFs burns ~50k+ tokens per transaction ($0.25+/query).**
- ☁ **Lost-in-the-Middle: Severe attention degradation in large contexts (e.g., missing 100% to 80% phase-downs).**
- ☁ **Transmitting full regulatory drafts outside client boundaries triggers governance hurdles.**

> "Sending entire documents to cloud models is expensive and error-prone. The solution is local deterministic diffing before cloud ingestion."

ARCHITECTURE

Automated Pipeline: Document Ingestion to Executive Synthesis

STEP 1

Ingestion Engine

pdf_parser.py strips running headers, disclaimers, and page numbers to prevent false diffs.

STEP 2

Deterministic Diffing

diff_engine.py segments statutory sections, isolating changes and bypassing the 90-95% unchanged baseline.

STEP 3

Local Refinement

Ollama worker (Llama 3.2) standardizes raw diff text into a typed, validated semantic JSON payload locally.

STEP 4 & TELEMETRY

Primary LLM Synthesis & SQLite Tracking

Primary LLM receives a lightweight ~1,000-token payload to generate the executive EY Client Impact Summary. All execution runs and token savings are logged to SQLite (contextsentry.db).

CODEBASE INFRASTRUCTURE

Modular Backend Infrastructure & Deliverables

pdf_parser.py

Layout-aware text extractor with bounding-box marginalia stripping.

diff_engine.py

Deterministic clause parser isolating MODIFIED, ADDED, and REMOVED sections.

worker.py

Two-stage worker (Ollama Llama 3.2 JSON cleaner + Primary LLM synthesizer).

telemetry.py

ContextSentry SQLite tracking engine logging token ROI in contextsentry.db.

server.py

Dual-interface serving Model Context Protocol (MCP) tool 'compare_regulatory_documents' and FastAPI REST API.

viewer.py

Interactive terminal diff inspector with section filtering (--section) and ANSI color diffs.

VERIFICATION & ROI

Automated Test Suite & Multi-Scale Financial Benchmarks

Robust Test Suite

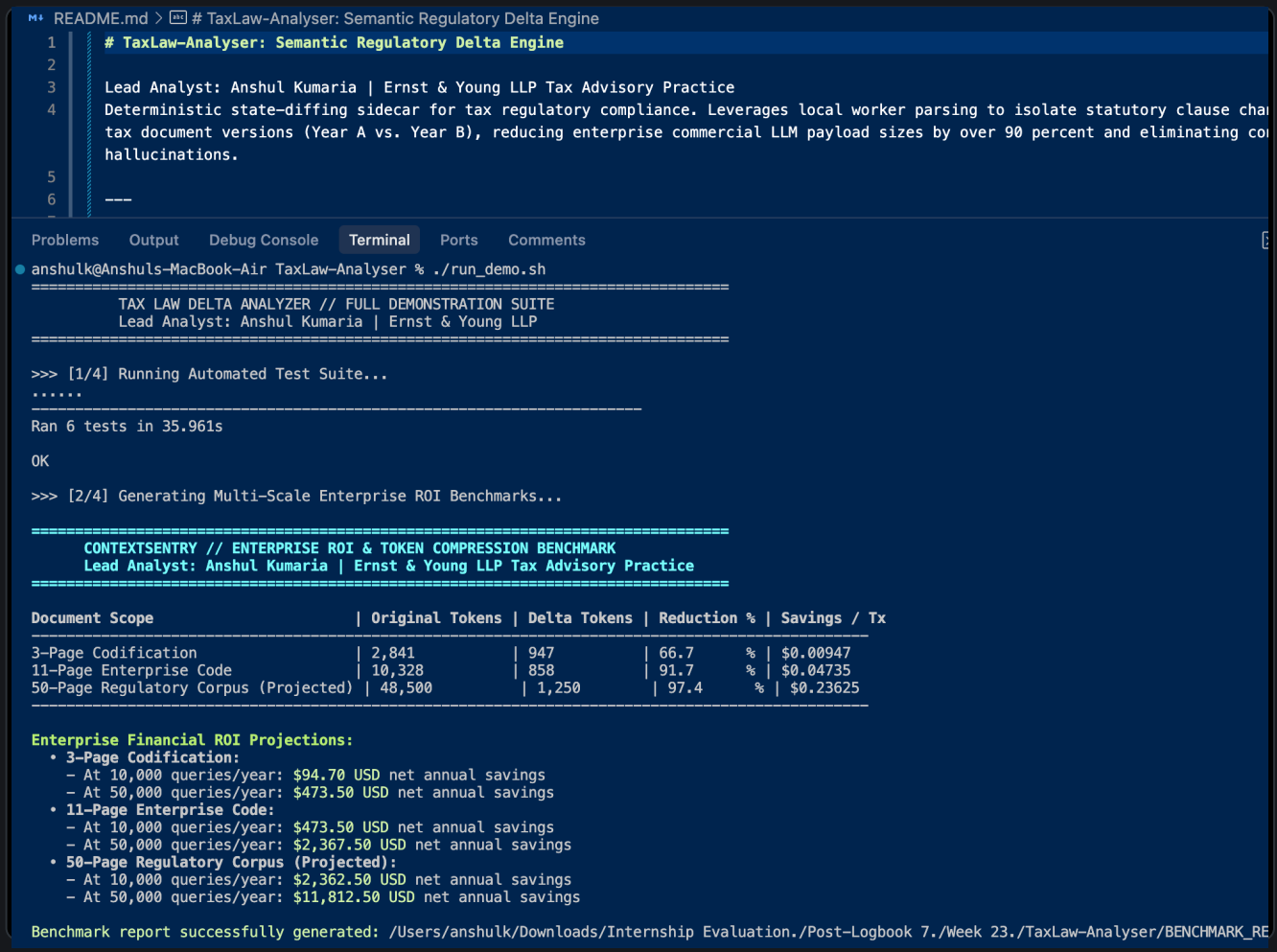
100% automated test pass (Ran 6 tests in 35.9s OK) covering marginalia filtering, diff calculations, and MCP protocol compliance.

Multi-Scale Token Savings

- 3-Page Codification: 66.7% Reduction ($0.00947 saved)
- 11-Page Enterprise Code: 91.7% Reduction ($0.04735 saved)
- 50-Page Regulatory Corpus: 97.4% Reduction ($0.23625 saved)

Firm Financial Impact

Over $11,812 USD annual net savings per practice group at 50k annual queries.



Caption: Terminal output showing unit test suite execution and multi-scale token compression benchmark matrix.

AUDIT TRAIL

Eliminating Noise: Deterministic Audit Trail

Marginalia Elimination

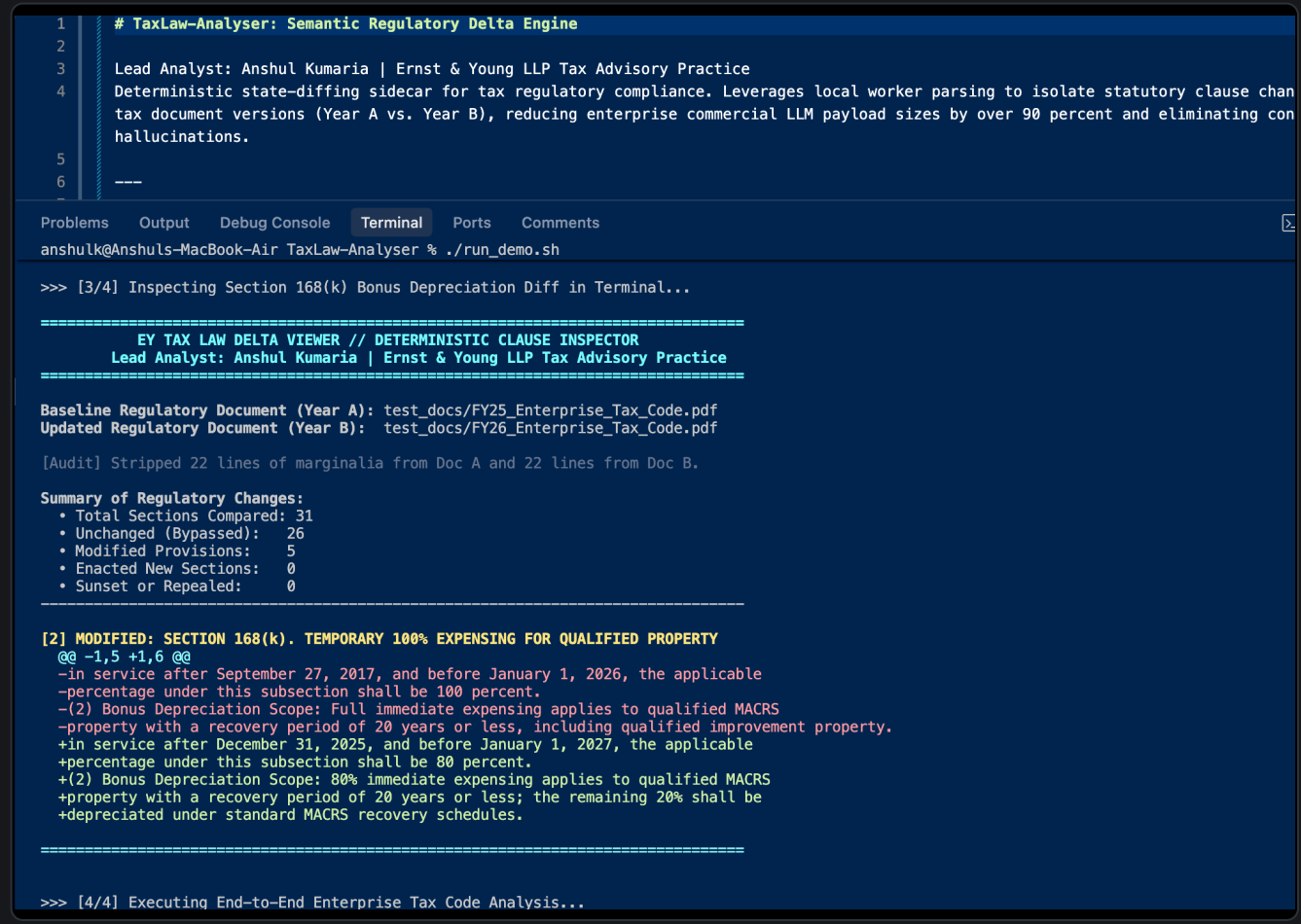
Stripped 22 lines of running headers, disclaimers, and page numbers per document, preventing false-positive diffs.

Section Isolation

Evaluated 31 statutory sections; 26 unchanged sections were identified and completely bypassed.

Section 168(k) Audit

Isolated the phase-down from 100% to 80% bonus depreciation for property placed in service after Dec 31, 2025.



Caption: viewer.py rendering Section 168(k) with red deletions (- 100%) and green additions (+ 80%) after stripping marginalia.

INGESTION & SEGMENTATION

Real-World Ingestion on 11-Page Enterprise Codifications

Engagement Target

Fortune 500 Enterprise Client codification analysis.

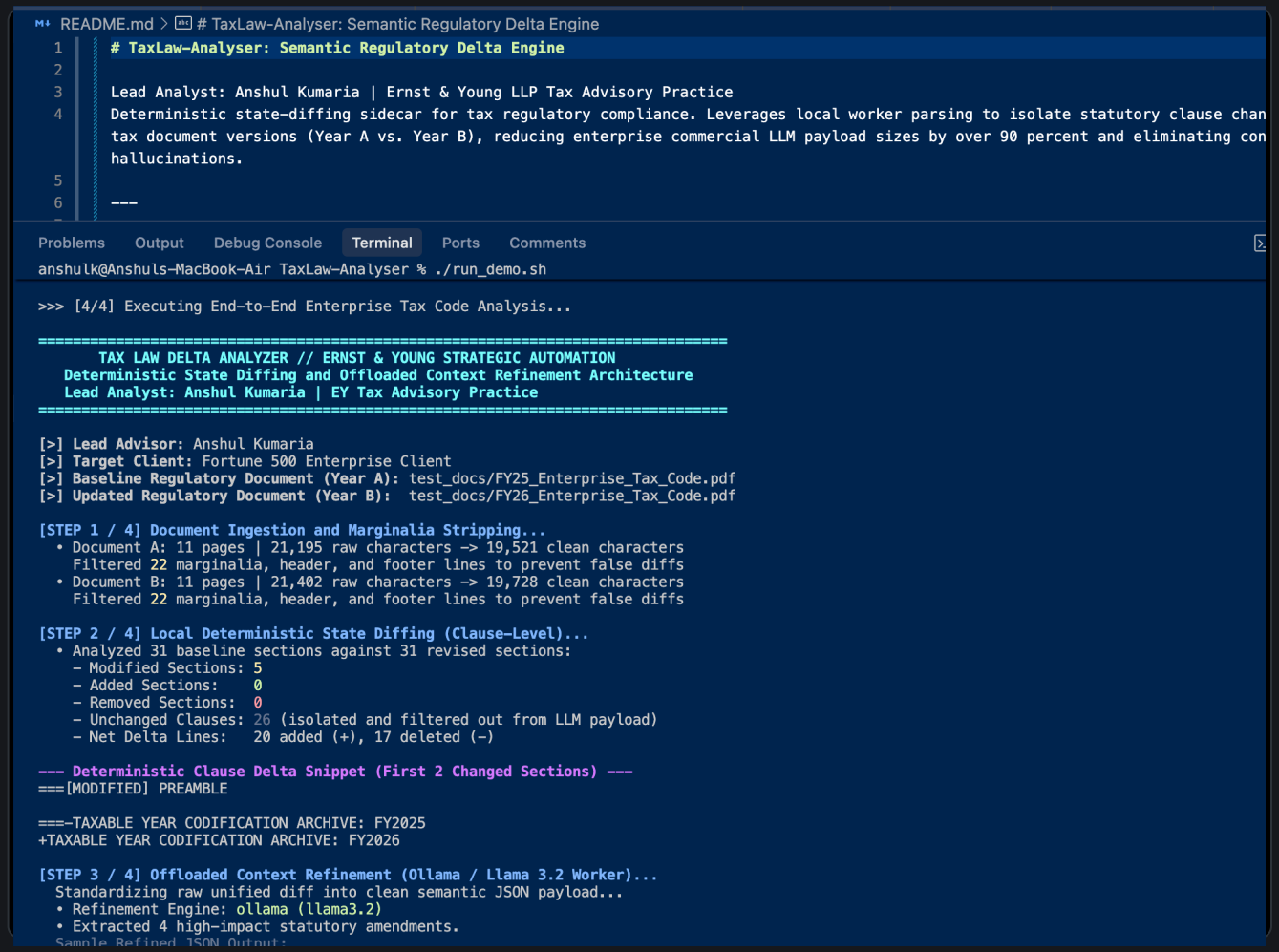
Document Ingestion Metrics

• Doc A: 11 pages | 21,195 chars → 19,521 clean chars (22 marginalia lines removed)

• Doc B: 11 pages | 21,402 chars → 19,728 clean chars (22 marginalia lines removed)

Deterministic Delta

5 modified sections isolated; 26 unchanged sections shielded from the LLM context window.



Caption: cli.py Step 1 & Step 2: Ingestion, layout-aware marginalia stripping, and clause-level diff segmentation.

CONTEXT REFINEMENT

Offloading Noise to Local LLM Workers

Local Data Sanitization

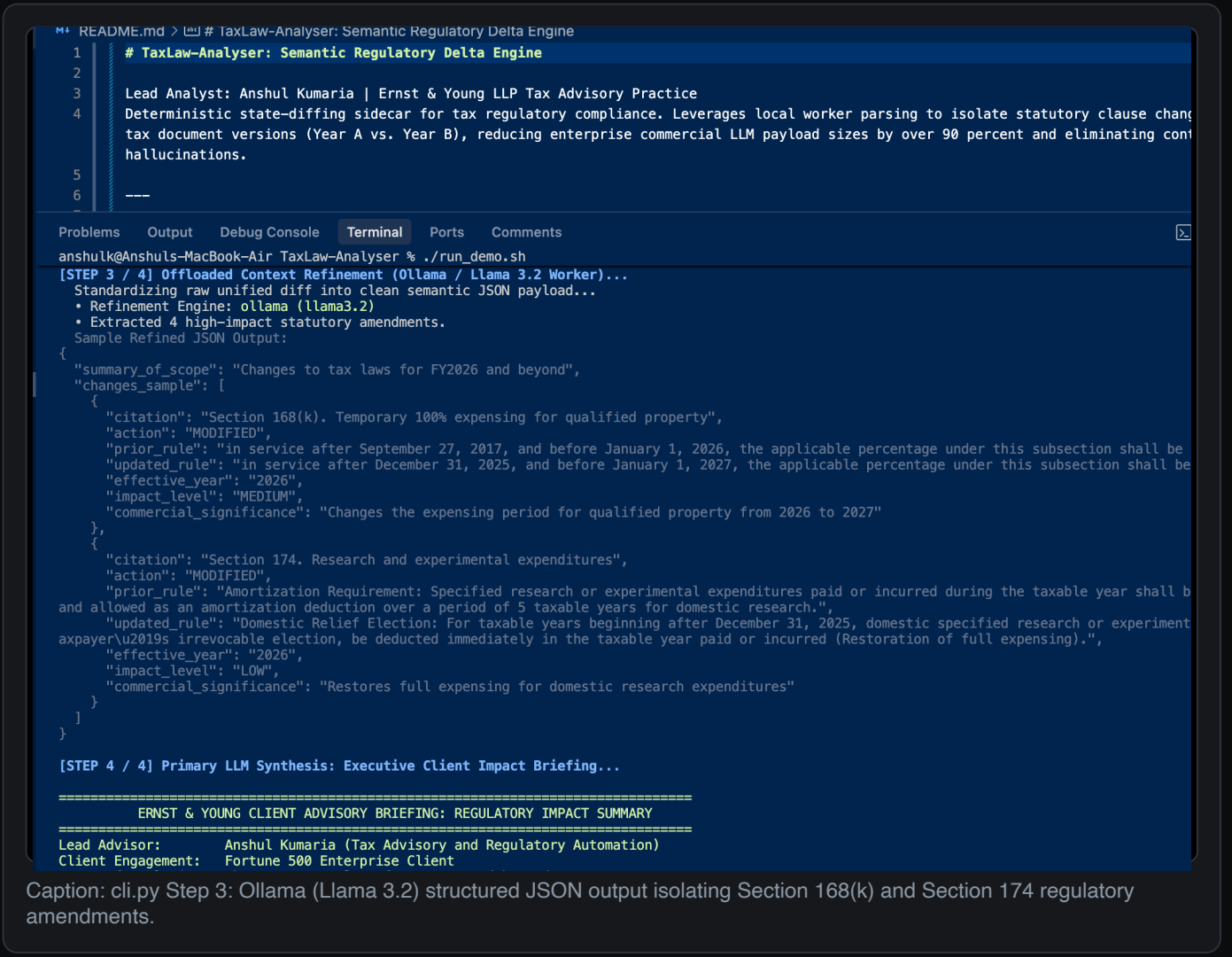
Raw diff syntax (+/- line diffs) is standardized on an on-premise Ollama worker running Llama 3.2.

Zero Cloud API Cost

Noise cleaning and JSON structuring happen locally at zero cloud compute cost.

Structured Semantic Output

Outputs strict JSON capturing statutory citation, prior rule, updated provision, effective taxable year, and impact level.



Caption: cli.py Step 3: Ollama (Llama 3.2) structured JSON output isolating Section 168(k) and Section 174 regulatory amendments.

CLIENT ADVISORY

Executive Client Advisory Briefing Generation

Lead Advisor

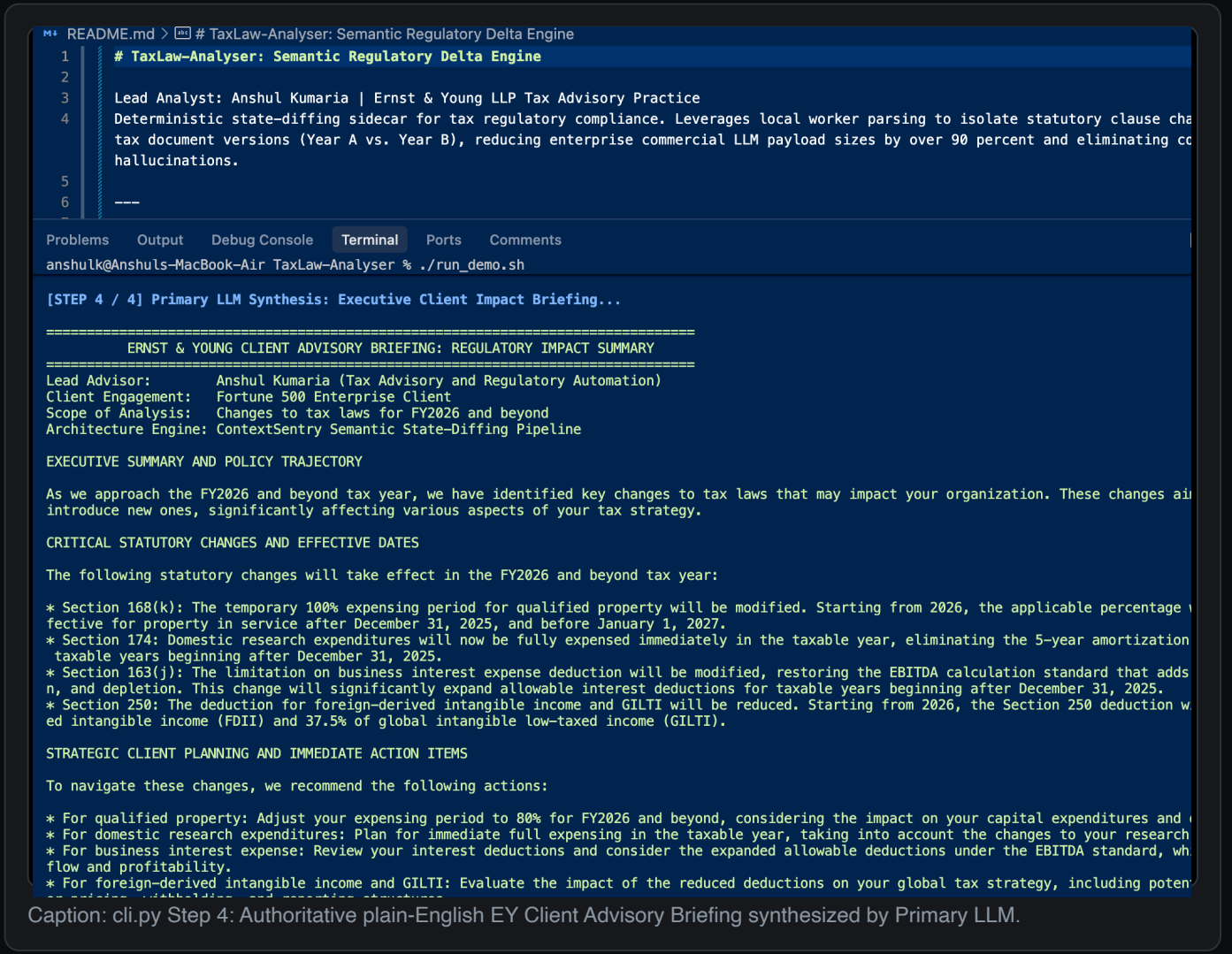
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Strategic Advisory Sections

• Section 1: Executive Summary & Macro Policy Trajectory
• Section 2: Statutory Amendments & Effective Dates (Sec 168k, 174, 163j, 250)
• Section 3: Strategic Planning Action Items (Capex acceleration, R&D election)

Professional Grade

Clean executive plain-text typography with zero markdown clutter.



Caption: cli.py Step 4: Authoritative plain-English EY Client Advisory Briefing synthesized by Primary LLM.

ACCOUNTING RISK & METRICS

Financial Accounting Risks & Empirical Token Savings

ASC 740 Financial Accounting Risks

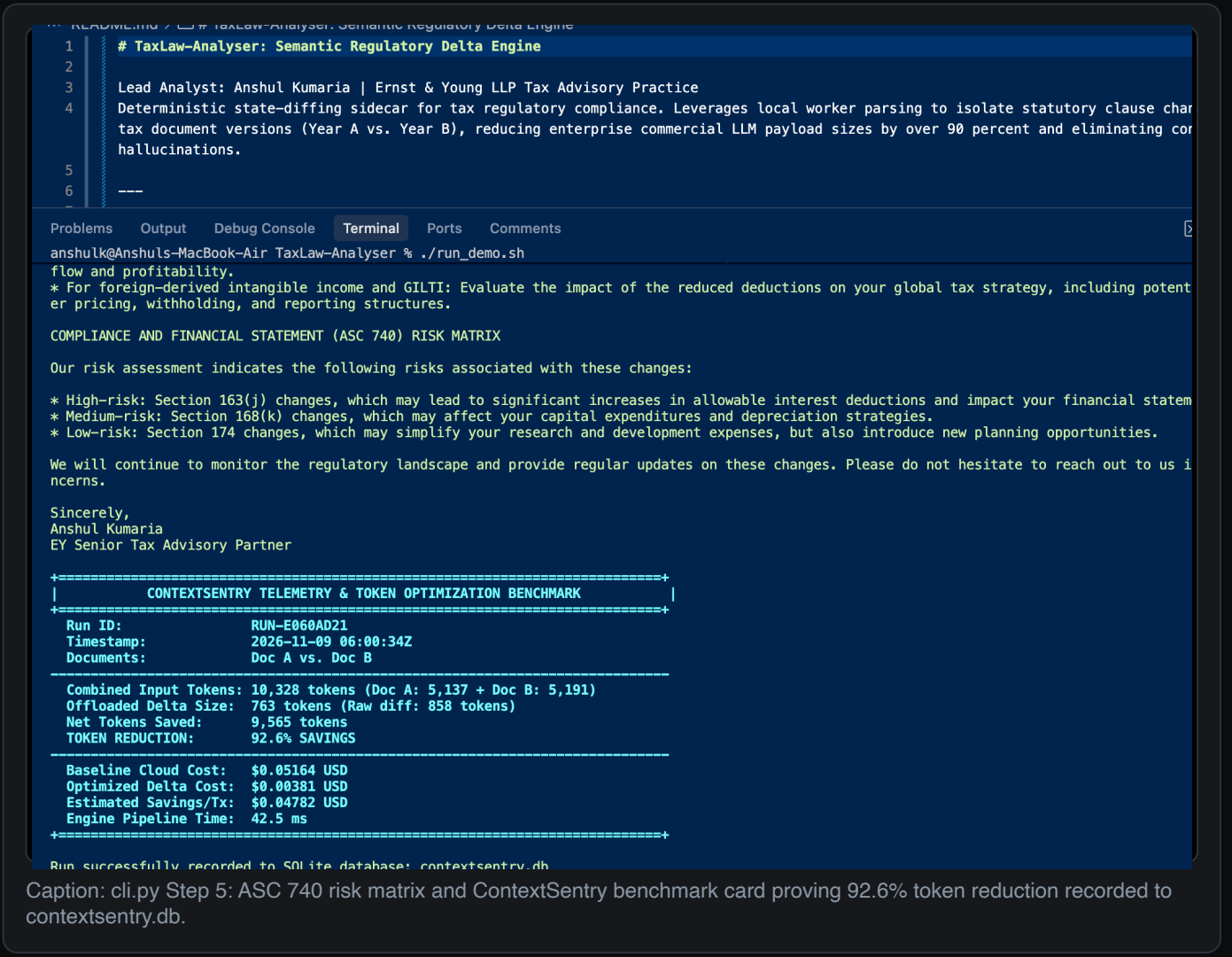
Identifies deferred tax asset/liability swings and discrete interim period reporting requirements.

Empirical Token Optimization

- Combined Size: 10,328 tokens | Offloaded: 763 tokens
- Net Saved: 9,565 tokens
- **EMPIRICAL TOKEN REDUCTION: 92.6% SAVINGS**

Cost & Latency

Baseline cloud cost $0.05164 → $0.00381 ($0.04782 saved/tx); pipeline latency: 42.5 milliseconds.



Caption: cli.py Step 5: ASC 740 risk matrix and ContextSentry benchmark card proving 92.6% token reduction recorded to contextsentry.db.

ENTERPRISE VALUE

Delivering Tangible Business Value to EY Tax Practices

TOKEN OPTIMIZATION — 92.6% SAVINGS

Payload reduced from 10,328 to 763 tokens; scales to 97.4% on 50-page regulatory corpora, saving over $11,800/year per practice group.

ZERO HALLUCINATIONS — 100% ATTENTION

Deterministic clause isolation guarantees zero missed sunset clauses or threshold changes ("Lost in the Middle").

CONSULTANT VELOCITY — 42.5 MILLISECONDS

Replaces 3+ hours of manual highlighter comparisons with an asynchronous pipeline executing in seconds.

SOVEREIGNTY & AUDITABILITY — SQLITE & MCP READY

Local worker keeps drafts sovereign; Model Context Protocol server connects seamlessly to AI agent ecosystems while logging every query in contextsentry.db.